Reg. No. | | | | | | | | | | | [12ME52/13ME52]

FIFTH SEMESTER B.E/B.TECH EXAMINATION, DECEMBER-2015

TURBOMACHINERY

Time: 3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full Questions.

1. a. Bring out the differences between turbomachines and positive Displacement machines. **4**

b. Define specific speed of pump. Show that specific speed of pump is given by $N_s = \frac{N\sqrt{Q}}{H^{3/4}}$. **4**

c. A turbine model of 1:10 develops 2.0 kw under a head of 6m at 500rpm. Find the power developed by the prototype under a head of 40 m. Also find the speed of the prototype and its specific speed. Assume the turbine efficiencies to remain same. **6**

2. a. With usual notations & velocity triangles derive alternate form of euler turbine equation and identify the components of energy transfer **6**

b. In an axial- flow turbine the discharge blade angles are 20° each, for both the stater and the rotor. The steam speed at the exit of the fixed blades is 140 m/s. the ratio $v_{vax}/u=0.7$ at the entry and 0.76 at the exit of the rotor blade. Find the inlet rotor blade angle, power developed for a mass flow rate of 2.6 kg/ s & degree of reaction R. **8**

3. a. Derive the expression for Head-capacity relation for a typical radial outward flow device assuming radial entry. Also draw the velocity triangles for :- (i) $\beta_2<90°$ (ii) $\beta_2=90°$ (iii) $\beta_2>90°$. **7**

b. In a mixed flow pump, fluid enters such that absolute velocity is axial at inlet and the relative velocity is radial at outlet. Evaluate degree of reaction and energy input to the fluid if the relative velocity at outlet is same as the tangential blade speed at inlet. The following data may be used:- ID=8cm ; OD=25cm, N= 3000rpm blade angle at inlet=45°. **7**

4. a. Define the following efficiencies with respect to hydraulic turbines;- **8**

i. Hydraulic efficiency

ii. Mechanical efficiency

iii. Volumetric efficiency

iv. overall efficiency

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b. In a power station, single jet Pelton wheel produces 23110 kw under a head of 1770 m while running at 750 rpm. Assume C_v=0.98, Φ=0.46 & overall efficiency=85%. Find ;- **6**

i. Jet diameter

ii. Wheel diameter

iii. No. of Buckets

5. a. Mention the functions of draft tube in a reaction hydraulic turbine. **3**

b. Draw a neat sketch of Francis turbine & explain its parts. **5**

c. A Kalpan turbine develops 10 Mw under an effective head of 8 m. the overall efficiency is 0.86, the speed ratio is 2.0, flow ratio is 0.6 the hub diameter is 0.35 times the outside diameter of the wheel. Find the diameter & speed of turbine. **6**

6. a. Prove that maximum blade efficiency with equiangular rotor blades for impulse steam turbine is $(n_b)_{max} = \cos^2 \alpha_1$ where, α_1= inlet nozzle angle. **6**

b. Steam issues from a nozzle of a De-lavel turbine with a velocity of 1200 m/s. nozzle angle is 20°. Blade speed is 400 m/s. Blades are symmetrical. Mass flow rate is 900 kg/hr calculate; **8**

i. Blades angles ii. Power developed iii. Blade efficiency iv. Draw the velocity triangles. Take Blade velocity co-efficient =0.8.

7. a. Obtain an expression for Minimum starting speed of a centrifugal pump. **5**

b. A 4- stage centrifugal pump has impeller each of 38 cm diameter and 1.9 cm wide at outlet, the outlet vane angle is 45° and the vanes occupy 10% of the outlet area. The manometric efficiency is 85% & overall efficiency is 75%. Determine the head generated by the pump when running at 1000 rpm, delivering 60 lps. Also find power required **6**

c. Explain the phenomena of cavitation in pumps. **3**

8. a. Draw the velocity triangles for an axial flow compressor and show that for an axial flow compressor having no axial thrust, degree of reaction is given by, **8**

$$R = \frac{v_a}{2u}\left[\frac{\tan\beta_1 + \tan\beta_2}{\tan\beta_1 + \tan\beta_2}\right]$$

where V_a= axial velocity

u=blade speed

β_1,β_2= inlet & outlet Blade angles.

b. Explain the following terms w.r.t compressor **6**

i. Surging ii. Choking iii. Slip.